Algorithmic Bias Audit Report

Target File: `./test/assign_team_leads_.py`

Audit Model: FATE Auditor GPT-4

Analysis Type: Static Code FATE Analysis

Status: 2 Flags Detected

Executive Summary

This report details instances of algorithmic bias identified within the source code during automated static analysis. The evaluation surfaced exclusionary logic based on protected attributes and/or proxies, which may propagate differential treatment through the Software Development Life Cycle (SDLC).

Detailed Findings

Finding 1: Proxy Bias

Association: POSITIVE

Non-Protected Attribute: leadership_style_match

Biased Code Snippet:

```
if candidate["leadership_style_match"] >= 0.9:
```

Issue Description: The function uses 'leadership_style_match' as a condition for a favorable outcome (selection as team lead). This score could be a proxy for protected attributes if the model defining the 'match' was trained on data reflecting historical biases in leadership roles (e.g., favoring specific genders or cultural backgrounds).

Impact Reasoning: Relies on a measurable attribute ('leadership_style_match') as a stand-in for a trait that correlates with a protected group, unfairly impacting groups where that attribute does not accurately indicate the intended outcome.

Finding 2: Threshold/Cutoff Bias

Association: POSITIVE

Non-Protected Attribute: leadership_style_match >= 0.9

Biased Code Snippet:

```
candidate["leadership_style_match"] >= 0.9
```

Issue Description: The code uses a fixed threshold of '0.9' on the 'leadership_style_match' score to determine eligibility for a team lead position. This rigid cutoff creates a binary decision where a small difference in score (e.g., 0.89 vs. 0.9) leads to a different outcome, potentially disadvantaging candidates near the boundary.

Impact Reasoning: Uses a fixed non-protected attribute ('leadership_style_match >= 0.9') with a threshold to determine eligibility or outcomes, where the rigid cutoff may produce biased treatment for individuals near the boundary.